# Capacity Fade of a Li-ion Cell Under Repeated Cycling

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This note models capacity retention of a lithium-ion cell as a geometric decay over cycle number. Battery capacity fade is summarised by a constant fractional loss per cycle.

Q(n) = Q0 (1 - d)^n

The cell has a rated capacity of 2.5 Ah and we use d = 0.0005 per cycle.
The cell was cycled for 1000 cycles at room temperature.
We observed 61% capacity retention after 1000 cycles in the model run.
End of life is defined as 80% capacity retention.

Limitation: the model assumes identical cycles and does not include calendar ageing or temperature effects.

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